

Cox Proportional Hazards Survival App

Output Artefacts: Reference Guide

MoveApps • github.com/meredithspalmer/MoveApps_CoxPropHazard

Introduction

This app implements a Cox Proportional Hazards (CoxPH) survival modelling workflow for animal tracking data loaded from MoveApps. It is designed to handle the realities of telemetry studies: staggered entry (animals collared at different times), right-censoring (animals still alive or lost to follow-up at study end), and small-sample separation (handled via automatic fallback to Firth's penalized regression).

The sections below describe every output artefact the app can produce — what it is, what it contains, how to read it, and why it matters. Artefacts are grouped by type: diagnostic summaries, core model outputs, and stratified group comparison outputs.

Note on Firth fallback: *When a covariate perfectly separates survivors from non-survivors, standard maximum-likelihood estimation fails to converge. The app detects this automatically and falls back to Firth's penalized Cox model. All artefacts clearly label which model was used.*

Note on optional features: *Several plots support two user-controlled options: `add_cis` (add 95% confidence interval ribbons) and `zoom_to_plot` (zoom the y-axis to the observed data range rather than displaying the full axis). When `zoom` is enabled, an annotation is added to the plot indicating the displayed range. These options apply to the covariate-means plots and all stratified KM and cumulative hazard plots.*

Section 1 — Data Diagnostic Artefacts

tracking_history.png

File	tracking_history.png
Type	Figure (PNG)
Always produced?	No — only when 'Tracking history plots' setting is enabled
Dimensions	Dynamic: scales with number of individuals and study duration

What it is

A horizontal Gantt-style chart showing the monitored period for each tagged individual. Each row is one animal; each grey bar spans its deployment period (from `deploy_on_timestamp` to `deploy_off_timestamp`); a blue point marks the start and a purple point marks the end. Upward-pointing arrows mark gaps in tracking longer than 30 days — periods where the individual was not being monitored (e.g., collar fell off and was later retrieved, or a second deployment occurred after a gap). The subtitle reports the total individual count, visible deployment count, and location count.

What it tells you

This figure provides an at-a-glance overview of your sample: how many individuals are in the study, when they entered and exited, how evenly they are distributed across the study period, and whether

any have data gaps worth noting. When running the analysis with survival years defined, the plot is built from the `yearly_survival` table; otherwise it uses the full `summary_table`.

How to interpret it

- Bars that end before the study period ends (especially with a purple endpoint) represent individuals whose deployment ended early — they may be deaths or censored events.
- Bars that extend to the very end of the study are likely censored survivors.
- Gap arrows indicate interrupted monitoring — these individuals contribute observation time in two separate windows.

Why it is valuable

Survival analyses assume you know when each individual entered and exited the risk period. This figure lets you visually confirm the data cleaning and time-windowing steps worked as intended, and flag any unusual deployment patterns before interpreting model results.

monthly_mortality.png

File	monthly_mortality.png
Type	Figure (PNG)
Always produced?	No — only when 'Monthly mortality plots' setting is enabled

What it is

A calendar heat map showing the number of confirmed mortality events per month per year. Months are columns, years are rows, and cell colour encodes mortality count using a PuBu (blue) palette (darker = more deaths). Months with zero deaths are shown in light grey. The subtitle reports the total mortality count and the full date range of the dataset.

What it tells you

This is a purely descriptive diagnostic. It summarises when deaths are happening in calendar time, revealing seasonality, year-to-year variability, and any suspicious clustering (e.g., a batch of deaths in one month that might be a data artefact or a discrete mortality event like a disease outbreak).

How to interpret it

- Consistent dark patches in the same months across years indicate seasonal mortality — common in species with winter die-offs or breeding-season predation.
- An isolated very dark cell in one year-month may reflect a mass mortality event or a data entry issue worth investigating.
- Many zero-months at the start or end of the time span reflect the ramp-up and wind-down of the study, not necessarily low mortality risk.

Why it is valuable

CoxPH models do not capture time-varying effects unless explicitly modelled. If you see strong seasonal clustering here and your covariates do not include a seasonal term, your model may be missing an important predictor.

Section 2 — Core Cox Model Artefacts

coxph_table.csv

File	coxph_table.csv
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Type	Table (CSV)
Always produced?	Yes

What it is

The primary numerical output of the multivariate Cox model. Each row corresponds to one coefficient — typically one row per non-reference level of each categorical covariate, or one row per continuous covariate. Columns are:

- term — the covariate level being described (e.g., sex: male relative to the reference female).
- estimate — the hazard ratio (HR): the multiplicative change in instantaneous mortality risk relative to the reference, holding all other covariates constant.
- conf.low / conf.high — lower and upper bounds of the 95% confidence interval around the HR.
- p.value — Wald test p-value (or profile-likelihood p-value for Firth's model).
- model — indicates whether 'Standard Cox' or 'Firth's Penalized Cox' was fitted.

How to interpret it

A hazard ratio of 1 means no difference from the reference group. $HR > 1$ means elevated risk; $HR < 1$ means reduced risk. A very wide confidence interval indicates high uncertainty, usually due to low event counts for that group.

Why it is valuable

This is the core inferential output of the app — it answers the primary biological question: which individual attributes are associated with elevated or reduced mortality risk, and by how much?

surv_at_means.csv

File	surv_at_means.csv
Type	Table (CSV)
Always produced?	No — only when 'Predicted survival at covariate means' is enabled

What it is

A tidy time-series table of the predicted survival function for a hypothetical "average" individual whose covariates are set to their means. Columns include time (days), estimate (survival probability), std.error, conf.low, conf.high, n.risk, n.event, and n.censor at each event time point.

Why it is valuable

The Cox model's baseline survival adjusted to covariate means gives a covariate-adjusted population summary — more useful for reporting than a simple KM curve. This table underlies surv_at_means_plot.png and can be used independently to extract specific survival probabilities.

surv_at_means_plot.png

File	surv_at_means_plot.png
Type	Figure (PNG)
Always produced?	No — only when 'Predicted survival at covariate means' is enabled
CI option	Confidence interval ribbon always shown
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled

What it is

A survival curve plot derived from `surv_at_means.csv`. The x-axis is time in days; the y-axis is survival probability from 0 to 100%. A shaded ribbon shows the 95% confidence interval. Below the main panel, a risk table shows the number of individuals still at risk and the cumulative number of events at regular time intervals. The subtitle labels which model type was used.

How to interpret it

The curve descends from 1.0 (all individuals alive at time zero) toward lower survival probabilities over time. A steep early drop indicates high early-period mortality. The confidence band widens at later time points as fewer individuals remain at risk.

Why it is valuable

This is the most intuitive communication tool the app produces. It translates abstract hazard ratio estimates into a tangible survival trajectory suitable for papers, reports, and presentations.

[cum_hazard_at_means_plot.png](#)

File	cum_hazard_at_means_plot.png
Type	Figure (PNG)
Always produced?	No — only when ‘Predicted survival at covariate means’ is enabled
CI option	Confidence interval ribbon always shown
Zoom option	y-axis zoomed to observed data range when <code>zoom_to_plot</code> is enabled

What it is

A plot of the Nelson-Aalen cumulative hazard function $H(t)$ for the average individual. The y-axis is cumulative hazard (unbounded above 0); the x-axis is time in days. A 95% confidence ribbon is shown. The subtitle labels which model type was used.

How to interpret it

$H(t)$ can be interpreted as the expected number of mortality events accumulated for a theoretical individual over $[0, t]$. It is related to survival by $S(t) = \exp(-H(t))$. Unlike the survival curve, the cumulative hazard is not bounded at 1, making it easier to distinguish mortality patterns that occur late in the study period.

Why it is valuable

The cumulative hazard scale is more sensitive to detecting non-proportional hazards visually, and it is the scale on which the PH assumption implies parallel lines between groups. It also supports intuitive risk communication: ‘over five years, a typical individual accumulates a cumulative hazard of 1.8, equivalent to a predicted survival of about 17%.’

[schoenfeld_residuals.png](#)

File	schoenfeld_residuals.png
Type	Figure (PNG) — one panel per covariate
Always produced?	No — only when ‘Schoenfeld residual plots’ is enabled
Note	Residuals are approximate when Firth’s model was used; a caption is added to the plot

What it is

A diagnostic plot for testing the proportional hazards (PH) assumption. One panel is produced per covariate. Each panel shows Schoenfeld residuals plotted against time, with a smoothed trend line and confidence band. The formal test used is the `cox.zph` (Grambsch-Therneau) test.

How to interpret it

- Random scatter around zero with a near-horizontal smoothed line: the PH assumption is satisfied. Proceed with standard HR interpretation.
- A clear upward or downward trend over time: the HR is changing over time, violating PH. The app will log a warning if `cox.zph` returns $p < 0.05$.
- A U-shaped or inverted-U pattern: the covariate effect is strongest mid-study.

Why it is valuable

The Schoenfeld residual plot is the standard diagnostic for the most important assumption of the CoxPH model. It should always be examined before reporting hazard ratios.

forest_combined.png

File	forest_combined.png
Type	Figure (PNG)
Always produced?	Yes, when at least one covariate is specified
Dimensions	Dynamic: panel heights scale proportionally to the number of terms per covariate

What it is

A combined forest plot showing hazard ratios and 95% confidence intervals for all covariates in a single figure. Each covariate occupies its own panel, stacked vertically using patchwork. Panel heights are proportional to the number of terms (levels) in each covariate, so a single-term continuous covariate occupies much less space than a multi-level categorical covariate. Within each panel, the covariate name and LR test result appear as a left-aligned title. Points are coloured red if $p < 0.05$ and blue if $p \geq 0.05$. The x-axis is on a log scale and is shared across all panels. A single legend appears below the bottom panel. The overall title and subtitle (identifying the model type) are added via patchwork `plot_annotation`.

How to interpret it

Points to the right of the dashed line ($HR = 1$) indicate elevated mortality risk relative to the reference group; points to the left indicate reduced risk. A confidence interval crossing $HR = 1$ indicates a non-significant result. The log scale is essential: equal distances in either direction represent equal multiplicative changes in risk.

Why it is valuable

Forest plots are the standard visualisation format for hazard ratio tables in ecological and medical literature. The combined layout allows rapid visual comparison across all covariates simultaneously in a single publication-ready figure, while variable panel heights prevent wasted whitespace for simpler covariates.

Section 3 — Stratified Group Comparison Artefacts

The following artefacts are produced once per grouping covariate (e.g., once for sex, once for `life_stage` if both are specified). In filenames below, `<cov>` is replaced with the covariate name

(e.g., sex, attachment_type). Artefacts marked 'categorical covariates only' are not produced for continuous (numeric) predictors such as weight.

model_tests_<cov>.csv

File	model_tests_<cov>.csv (e.g., model_tests_sex.csv)
Type	Table (CSV)
Always produced?	Yes, for each active covariate

What it is

A table of three omnibus test statistics for the univariable CoxPH model for this covariate: the likelihood ratio test (LRT), the Wald test, and the score test. Columns are: covariate, test, chi_sq, df, p_value, and note (indicates if Firth's was used).

How to interpret it

All three tests evaluate whether the covariate has any overall effect on survival (null: all group HRs = 1). A small p-value indicates that at least one group differs. The LRT p-value is reported in the KM plot subtitle and is the recommended summary statistic. If the three tests give substantially different p-values, this signals a small sample or model instability.

Why it is valuable

The omnibus test answers the prior question: is there any evidence at all that this covariate matters? This should be checked before interpreting pairwise comparisons.

cox_hr_<cov>.csv

File	cox_hr_<cov>.csv (e.g., cox_hr_sex.csv)
Type	Table (CSV)
Always produced?	Yes, for each active covariate

What it is

A tidy hazard ratio table from a CoxPH model fit with only the specified grouping covariate as the predictor — a univariable (unadjusted) model for that covariate. Columns include term, estimate (HR), conf.low, conf.high, p.value, and covariate. For continuous covariates the HR represents the change in risk per one-unit increase.

How it differs from coxph_table.csv

The main coxph_table.csv is from the multivariate model and shows adjusted hazard ratios. The cox_hr_<cov>.csv files are from single-covariate models and show unadjusted hazard ratios. The two may differ substantially if covariates are correlated.

Why it is valuable

Univariable results are often required in supplementary tables for ecological publications, and they provide a sanity check: if unadjusted and adjusted HRs differ dramatically, this signals confounding that may need to be addressed in interpretation.

pairwise_<cov>.csv

File	pairwise_<cov>.csv (e.g., pairwise_model.csv)
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Type	Table (CSV)
Always produced?	Only when the covariate has more than 2 groups (categorical covariates only)

What it is

A table of pairwise hazard ratio comparisons between every pair of groups, using a separate two-group CoxPH model for each pair. Columns are: group_1, group_2, hr, p_wald (raw Wald p-value), p_bonferroni (Bonferroni-corrected p-value), significant (Yes/No at alpha = 0.05), and method (Standard Cox or Firth).

How to interpret it

An omnibus test that rejects the null only tells you some groups differ — not which ones. This table identifies exactly which pairs are significantly different after Bonferroni correction. Note that Bonferroni correction is conservative and may produce false negatives with many groups; treat this table as a screening tool.

Why it is valuable

When comparing three or more groups (e.g., juvenile, sub-adult, adult), researchers need to know which specific pairwise transitions are significant. This table provides those comparisons in a rigorous, multiple-testing-corrected format.

km_by_<cov>.png

File	km_by_<cov>.png (e.g., km_by_sex.png)
Type	Figure (PNG)
Always produced?	Yes, for each active categorical covariate
CI option	95% confidence interval ribbons shown when add_cis is enabled
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled

What it is

A Kaplan-Meier (KM) survival curve with one line per group within the covariate, coloured using the Viridis “turbo” palette for accessibility. The subtitle reports the Cox likelihood ratio test chi-square statistic and p-value. When add_cis is enabled, shaded ribbons show 95% confidence intervals. When zoom_to_plot is enabled, the y-axis is restricted to the observed survival range and an annotation marks the zoom bounds.

What Kaplan-Meier estimation is

The KM estimator is a non-parametric method that estimates the survival function directly from the data without model assumptions. It naturally handles right-censoring by removing censored individuals from the at-risk set before the next event.

How to interpret it

Each step-down in a line represents one or more deaths. If one group's line sits consistently lower, that group has higher mortality risk. Groups whose confidence intervals overlap substantially do not show statistically distinguishable differences. The number of individuals at risk at each time point is critical context for interpreting confidence interval width.

Why it is valuable

The KM plot is the most widely recognised way to visualise survival differences between groups and is expected in virtually all ecological survival analysis publications. Unlike the Cox model, it makes no assumptions about the shape of the hazard function.

km_summary_<cov>.csv

File	km_summary_<cov>.csv (e.g., km_summary_sex.csv)
Type	Table (CSV)
Always produced?	Yes, for each active categorical covariate

What it is

A per-group summary table from the KM fit. Columns are: strata (group label), n_risk_start (sample size at time zero), n_events (total deaths observed), median_surv (time at which 50% of the group had died, or NA if fewer than 50% died), surv_at_1yr (KM-estimated survival probability at 365 days, or NA if follow-up is shorter), lr_p_value, and covariate.

Why it is valuable

This table provides the key statistics needed for a results section: 'Females had a one-year survival of 0.72 (n=34), compared to 0.58 for males (n=29); Cox LR p = 0.04.' It is formatted for direct extraction into manuscripts and reports.

cumhaz_by_<cov>.png

File	cumhaz_by_<cov>.png (e.g., cumhaz_by_sex.png)
Type	Figure (PNG)
Always produced?	Yes, for each active categorical covariate
CI option	95% confidence interval ribbons shown when add_cis is enabled
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled

What it is

Stratified cumulative hazard curves, one per group. The x-axis is time in days; the y-axis is cumulative hazard $H(t)$. The subtitle notes that parallel lines support the proportional hazards assumption. When add_cis is enabled, confidence ribbons are shown. When zoom_to_plot is enabled, the y-axis ceiling is set to the observed maximum and annotated.

How to interpret it

- Parallel, non-crossing lines: PH assumption holds. The Cox HR estimate is reliable for that covariate.
- Lines that converge or cross: PH assumption is likely violated.
- Lines that start parallel and then fan apart: early survival differences become more pronounced over time.

Why it is valuable

This is a visual complement to the Schoenfeld residuals test. The prediction under PH (parallel lines) is geometrically obvious, making this plot accessible to audiences unfamiliar with residual-based diagnostics. Using both diagnostics together provides a more complete picture of whether the PH assumption holds.

annual_surv_by_<cov>.png and annual_surv_<cov>.csv

File (figure)	annual_surv_by_<cov>.png (e.g., annual_surv_by_sex.png)
File (table)	annual_surv_<cov>.csv (e.g., annual_surv_sex.csv)
Type	Figure (PNG) + Table (CSV)
Always produced?	Only when survival years are defined; categorical covariates only
Dimensions	Dynamic: width and height scale with number of groups and years

What it is

A grouped bar chart and its underlying data table showing the empirical annual survival rate per group per user-defined survival year. Bars are coloured by group using the Viridis “turbo” palette; sample size labels (n=XX) are printed in white inside each bar near the top. Bar width and plot dimensions scale dynamically with the number of groups and years. The table contains columns: survival_year, the covariate column, n_animals, n_deaths, and surv_rate.

How it is calculated

Annual survival rate is computed as $1 - (n_deaths / n_animals)$ within each group-year stratum. This is a simple cohort proportion, not a KM estimate, and does not account for staggered entry or censoring within the year. It should be interpreted as a descriptive summary of observed annual mortality rates, not a formal survival estimate.

How to interpret it

- Year-to-year variability within a group (bars of the same colour varying in height) reflects real or stochastic differences in annual mortality.
- Consistent differences between groups across all years suggest a robust group effect on survival.
- Group differences that reverse in some years suggest an interaction between group identity and year.
- Very short bars (low survival rates) with small n labels (e.g., n=1 or n=2) should be interpreted with extreme caution — single-death estimates are unreliable.

Why it is valuable

Long-term wildlife studies often need to track how annual survival rates change over time and whether group differences are stable across years. This plot makes year-specific patterns visible in a way that a single overall KM curve cannot.